SIMATS ENGINEERING

ASSESSMENT TOOL 3 – ANSWERED

Course: Artificial Intelligence | Course Code: CSA1708

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.

Question 1 – Intelligent Agents in a Smart Home Automation System

1. Introduction

An intelligent agent perceives its environment through sensors and acts through actuators to achieve its objectives. A smart home senses motion, temperature and time, and controls lights, heating/cooling and security systems.

PEAS	Smart home system
Performance measure	Occupant comfort, low energy consumption, home security and fast response.
Environment	Rooms, occupants, weather, time of day, doors and windows.
Actuators	Lights, heating/cooling unit, locks, alarm and notifications.
Sensors	Motion detectors, temperature sensors, light sensors, clock, door/window contacts and cameras.

Environment characteristics: Partially observable, stochastic, dynamic, sequential and continuous.

2. Types of Intelligent Agents

Agent type	How it works	Example	Limitation
Simple reflex	Uses current percept with if–then rules; no memory.	If motion is detected at night, turn on the light.	Fails with partial observability; cannot learn or plan.
Model-based reflex	Maintains an internal state/model from percept history and applies rules.	Remembers room occupancy and tracks window/temperature status.	Still rule-based; no explicit goal or comparison of alte
Goal-based	Uses a model and explicit goals; plans/searches for actions that achieve them.	Plan AC operation to make the room 24°C by arrival; secure the house before bedtime.	Cannot rank different successful ways by cost or com
Utility-based	Assigns utility to outcomes and chooses the highest expected utility.	Balance comfort, energy cost and security; prefer cheaper off-peak cooling.	Needs a good utility function and more computation.
Learning agent	Learns from experience using learning, critic, performance and problem-generator components.	Learns preferred temperature and lighting patterns over time.	Needs time/data; bad feedback can cause wrong bel

3. Most Effective Agent – Hybrid Approach

A hybrid layered agent is most effective because no single agent type is sufficient. Reflex rules provide fast safety responses; a model-based layer tracks state; a goal-based layer plans; a utility-based layer balances comfort, energy and security; and a learning component personalizes behaviour.

Layer	Agent type	Role
1 – Safety	Simple reflex	Immediate alarm/unlock/notification response; motion in darkness → light on.
2 – State tracking	Model-based	Maintains occupancy, doors/windows, temperature trend and time model.
3 – Planning	Goal-based	Schedules heating/cooling, lighting and security routines.
4 – Optimisation	Utility-based	Chooses actions that maximise comfort, energy efficiency and security.
Learning	Learning agent	Updates preferences and utility weights from user feedback.

Conclusion: The hybrid agent gives the best balance of comfort, energy efficiency and security in a changing, partially observable smart-home environment.

Question 2 – Robot in an Unknown Maze: UCS and IDS

1. Problem formulation

States: discovered maze cells/junctions. Actions: move forward, left, right or back when not blocked. Step cost: distance, time or energy. Goal test: robot reaches the exit. Solution: sequence of moves from start to exit, ideally with minimum cost.

Because the maze layout is unknown and there is no heuristic, the robot uses uninformed (blind) search and builds its map while moving.

2. Uniform Cost Search (UCS)

UCS keeps a priority queue ordered by path cost g(n). It expands the lowest-cost node first and replaces an older path when a cheaper path is found. Therefore, UCS finds the minimum-cost route when step costs are non-negative. If all steps have equal cost, UCS behaves like Breadth-First Search.

3. Iterative Deepening Search (IDS)

IDS performs depth-limited DFS with limits 0, 1, 2, 3, … until the goal is found. It combines the completeness and shallowest-solution property of BFS with the low memory usage of DFS. It is optimal only when all step costs are equal.

4. UCS vs IDS

Property	UCS	IDS
Complete?	Yes, if every step cost ≥ ε > 0.	Yes, if branching factor is finite.
Optimal?	Yes, for non-negative step costs.	Yes only when all step costs are equal.
Time	O(b^(1+■C*/ε■))	O(b^d)
Space	O(b^(1+■C*/ε■))	O(b·d)
Main advantage	Finds cheapest path with varying costs.	Very low memory; no depth limit to guess.
Main disadvantage	Very high memory; can be slow.	Repeats work and ignores varying step costs.

5. Intelligent Agent Types in the Maze

Agent	Behaviour / suitability
Simple reflex	Acts from the current percept only; may loop forever in mazes with cycles.
Model-based	Stores the discovered map and visited cells; necessary for limited sensing and avoiding loops.
Goal-based	Uses the goal ‘reach the exit’ and search such as UCS/IDS to plan moves.
Utility-based	Balances distance, time, energy and risk; especially useful when costs vary.
Learning	Uses previous runs to remember dead ends and improve future performance.

6. Best combination and justification

Recommended: a model-based, goal-based agent using IDS for exploration together with UCS for cost-optimal planning on the discovered map.

- Model-based: stores the discovered map and visited cells to avoid loops.
- Goal-based: searches deliberately for the exit.
- IDS: suitable when exit depth is unknown and memory is limited; uses O(b·d) memory.
- UCS: finds the minimum-cost route when corridor costs differ. If all moves cost the same, IDS is sufficient and uses less memory.

Final conclusion: UCS provides cost optimality and IDS provides memory efficiency. Combined with model-based and goal-based design, the robot can explore an unknown maze, remember what it has seen and follow the cheapest route to the exit.